

**Laxmi Charitable Trust's**  
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**  
**Department of Information Technology (B.Sc.I.T Semester V)**  
**Dot .Net Core and Progressive Web Application Practical**

## Practical – X

|                                |                                     |
|--------------------------------|-------------------------------------|
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| Class: TYIT                    | Batch: 1                            |
| Date of Assignment: 22/09/2026 | Date/Time of Submission: 22/09/2026 |

### A. Create a PWA With a Web App manifest file Including.

#### Code:

##### (manifest.json):

```
{
  "name": "My PWA Application",
  "short_name": "MyPWA",
  "start_url": "/index.html",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "447x447",
      "type": "image/png"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "447x447",
      "type": "image/png"
    }
  ]
}
```

##### index.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>My PWA</title>
  <link rel="manifest" href="manifest.json">
  <meta name="theme-color" content="#0d6efd">
</head>
<body>
  <h1>My Progressive Web App</h1>
  <button onclick="showNotification()">
    Show Notification
  </button>
  <script src="app.js"></script>
</body>
</html>
```

#### app.js:

```
if ('serviceWorker' in navigator)
{
  navigator.serviceWorker.register('service-worker.js')
    .then(() => console.log('Service Worker Registered'));
}
function showNotification() {
}
Notification.requestPermission().then(permission =>
{
  if (permission === 'granted')
  {
    navigator.serviceWorker.ready.then(registration =>
    {
      registration.showNotification('PWA Notification', {
        body: 'Push Notification Working Successfully!',
        icon: 'icons/icon1.png'
      });
    });
  }
});
```

#### service-worker.js:

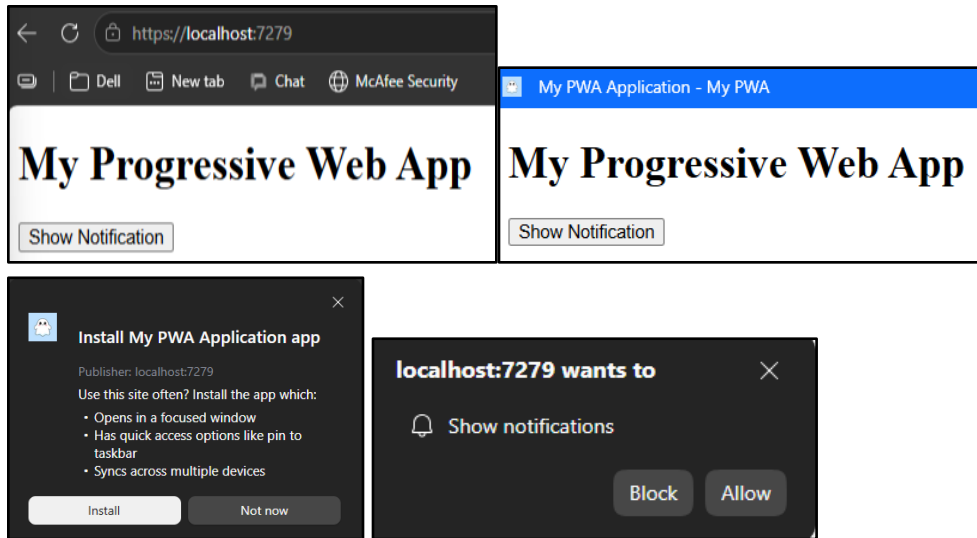
```
self.addEventListener('install', event =>
{
  console.log('Service Worker Installed');
});

self.addEventListener('activate', event =>
{
  console.log('Service Worker Activated');
});
```

#### program.cs:

```
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();
app.UseDefaultFiles();
app.UseStaticFiles();
app.Run();
```

Output:



## B) Develop an E-Commerce PWA with product listing and offline cart functionality.

Code:

Index.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>E-Commerce PWA</title>
  <link rel="manifest" href="manifest.json">
  <meta name="theme-color" content="#0d6efd">
</head>
<body>
  <div class="container">
    <div class="cart-section">
      <h2>Cart Items</h2>
      <ul id="cartList"></ul>
    </div>
    <div class="product-section">
      <h2>Products</h2>
      <div class="products">
        <div class="product">
          <h3>Laptop</h3>
          <p>Price: ₹ 50000</p>
          <button onclick="addToCart('Laptop')">Add
            to Cart</button>
        </div>
        <div class="product">
          <h3>Mobile</h3>
          <p>Price: ₹ 20000</p>
          <button onclick="addToCart('Mobile')">Add
            to Cart</button>
        </div>
        <div class="product">
          <h3>Headphones</h3>
          <p>Price: ₹ 3000</p>
          <button
            onclick="addToCart('Headphones')">Add To
            Cart</button>
        </div>
      </div>
    </div>
  </div>
</body>
</html>
```

```
</head>
<body>
  <h1>E-Commerce PWA</h1>
  <div class="container">
    <!-- Cart Section -->
    <div class="cart-section">
      <h2>Cart Items</h2>
      <ul id="cartList"></ul>
    </div>
    <!-- Product Section -->
    <div class="product-section">
      <h2>Products</h2>
      <div class="products">
        <div class="product">
          <h3>Laptop</h3>
          <p>Price: ₹ 50000</p>
          <button onclick="addToCart('Laptop')">Add
            to Cart</button>
        </div>
        <div class="product">
          <h3>Mobile</h3>
          <p>Price: ₹ 20000</p>
          <button onclick="addToCart('Mobile')">Add
            to Cart</button>
        </div>
        <div class="product">
          <h3>Headphones</h3>
          <p>Price: ₹ 3000</p>
          <button
            onclick="addToCart('Headphones')">Add To
            Cart</button>
        </div>
      </div>
    </div>
  </div>
</body>
</html>
app.js:
```

```
if ("serviceWorker" in navigator) {
  navigator.serviceWorker.register("service-worker.js")
    .then(() => console.log("Service Worker
Registered"));
}
let cart = JSON.parse(localStorage.getItem("cart")) || [];
displayCart();
function addToCart(product) {
  cart.push(product);
  localStorage.setItem("cart", JSON.stringify(cart));
  displayCart();
}
function displayCart() {
  let cartList = document.getElementById("cartList");

  cartList.innerHTML = "";

  cart.forEach(item => {
    let li = document.createElement("li");

    li.textContent = item;

    cartList.appendChild(li);
  });
}
```

#### service-worker.js

```
const CACHE_NAME = 'ecommerce-cache-v1';
const urlsToCache = [
  '/index.html',
  '/app.js',
  '/manifest.json'
];
self.addEventListener('install', event => {
  event.waitUntil(
    caches.open(CACHE_NAME)
      .then(cache => {
        return cache.addAll(urlsToCache);
      })
  );
});
```

```
self.addEventListener('fetch', event => {
  event.respondWith(
    caches.match(event.request)
      .then(response => {
        return response || fetch(event.request);
      })
  );
});
```

#### program.cs:

```
var builder = WebApplication.CreateBuilder(args);
var app = builder.Build();
```

```
app.UseDefaultFiles();
app.UseStaticFiles();
```

```
app.Run();
```

#### manifest.json:

```
{
  "name": "E-Commerce PWA",
  "short_name": "ShopPWA",
  "start_url": "/index.html",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "128x128",
      "type": "image/png"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
```

#### Output:

